Koa Shen

Mechanical Engineer
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EDUCATION

University of California, Santa Barbara — Expected June 2028
B.S. Mechanical Engineering | Major GPA: 3.97 | Cumulative GPA: 3.94 — Santa Barbara, CA

- Dean's Honors every quarter. Coursework: Mechanics of Materials, Statics, Dynamics, Physics, MATLAB.

EXPERIENCE

Fabrication Team Lead — Fall 2024 - Present
Gaucho Racing, Formula SAE Electric | UC Santa Barbara — Santa Barbara, CA

- Secured $2,000 in undergraduate research funding for steering-wheel R&D; used hand calculations and FEA to compare CFRP, GFRP, 6061-T6, and 7075-T6 aluminum for a lightweight baseplate.
- Tested the selected design for ergonomics, strength, and FSAE compliance; the GR25 redesign cut assembly weight 25% and cost 30%, and later FEA-guided iterations reduced mass from 5 lb to 1.7 lb.
- Led 15 teammates through GR26 driver-safety systems, including the team's first composite seat (about 4 lb), establishing a baseline for material selection, mold and layup process, fit, and installation.
- CNC-machined 33 suspension parts from 4130 steel and 7075-T6 aluminum, holding bearing interfaces within 0.0005 in; traced upright scrap to part flex under machining loads that pushed bores out of tolerance.

Undergraduate Researcher — Fall 2025 - Present
Hawkes Lab, UC Santa Barbara — Santa Barbara, CA

- Designed and 3D-printed custom molds with complex curvature for composite parts and prepared them for layup.
- Fabricated parts using wet layup and prepreg across different weaves and composite fabrics; experimented with vacuum-bagging setups.
- Iterated on mold design to improve part stiffness, evaluating results through hands-on empirical testing.

Maintenance Technician — Summer 2025
Exploratorium — San Francisco, CA

- Traced acrylic enclosure stress fractures to over-constraining discrete fasteners; designed a load-spreading bezel, improved the bolting pattern, and sealed cracks with acrylic solvent and silicone paste, avoiding $3,000+ in replacement costs.
- Addressed membrane wear and enclosure-seal leaks in a piston-crank exhibit that demonstrates the non-Newtonian behavior of oobleck.
- Redesigned aluminum encoder mounts worn by garnet-sand ingress with a thicker flange and toleranced fit.

Mechanical Engineering Intern — Summer 2026 - Present
SLAC National Accelerator Laboratory, LCLS — Menlo Park, CA (part-time remote)

- Performed statics and dynamics hand calculations to qualify stages and mounts against crash and drop load cases protecting $100k+ detectors.
- Redesigned optics packaging with kinematic mounts, alignment features, and detector strain relief; released Solid Edge designs with GD&T through Teamcenter PDM.

PROJECTS

Closed-Loop Cycloidal Actuator | Personal Project — Summer 2026 - Present

- Printed reducer components in PETG and ABS on a Bambu P1S, orienting layers to maximize radial load capacity in the housing's primary load case while holding the tolerances that govern backlash and friction.
- Packaged a 15:1 reduction within a NEMA 17 motor's 42 mm square face; bench-validated encoder-based closed-loop position correction in C++ firmware.

TECHNICAL SKILLS

Materials: CFRP, GFRP, wet layup, prepreg, vacuum bagging, 4130 steel, 6061-T6/7075-T6 aluminum, PETG, ABS
Analysis: Mechanics of materials, hand calculations, FEA, statics and dynamics, failure analysis, material selection
Design and Fabrication: SolidWorks, Solid Edge, GD&T, DFM, CNC/manual machining, welding, 3D printing
Software: Python, MATLAB, C/C++, Git